■ Developer Mock Test Results

Test ID:	be044bf3-8e65-4d...	**Student ID:**	1409
Test Type:	Developer Assessment	**Date:**	2026-03-02 16:49:04
Overall Score:	0/25 (0.0%)	**Performance:**	■■ Needs Improvement
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	0/10	0.0%	■■ Needs Work
■ MCQ/Theory	0/10	0.0%	■■ Needs Work
■ Coding	0/5	0.0%	■■ Needs Work

■ Detailed Question Review

■ APTITUDE SECTION (0/10 - 0.0%)

■ Q1. If 5 men can do a piece of work in 10 days, how many men would be required to do the same work in 5 days?

Your Answer: __SKIPPED__
Correct Answer: 10
■ *To find the number of men required to do the work in 5 days, we can use the formula: men * days = constant. Given that 5 men can do the work in 10 days, we have 5 * 10 = 50 man-days. To do the same work in 5 days, we need 50 man-days / 5 days = 10 men. The correct answer is 10, which is option B. The user likely skipped the calculation step, which is essential to solving this problem.*

■ Q2. If 2 men can do a piece of work in 6 days, how many men would be required to do the same work in 3 days?

Your Answer: __SKIPPED__
Correct Answer: 6
■ *To find the number of men required to do the work in 3 days, we can use the formula: (number of men) * (number of days) = constant. Given that 2 men can do the work in 6 days, we have 2 * 6 = 12. To do the same work in 3 days, we need to find the number of men such that (number of men) * 3 = 12. The key calculation step is: number of men = 12 / 3 = 4. However, the correct answer is stated as 6, but based on the calculation, the correct answer should be 4. The likely mistake in the given correct answer is the incorrect calculation or misunderstanding of the formula.*

■ Q3. If a product is marked at a 20% discount, and then a 10% tax is added, what is the final price of the product if the original price is Rs.100?

Your Answer: __SKIPPED__
Correct Answer: Rs.108

■ *To find the final price, first calculate the discount: 20% of Rs.100 is 0.20 * 100 = Rs.20, so the price after discount is 100 - 20 = Rs.80. Then, calculate the 10% tax on the discounted price: 0.10 * 80 = Rs.8, so the final price is 80 + 8 = Rs.88. The correct answer is not among the options, the calculation yields Rs.88, not Rs.108, suggesting a mistake in the given correct answer.*

■ Q4. If all A are B, and all B are C, what can be concluded?

Your Answer: __SKIPPED__
Correct Answer: All A are C
■ *The correct answer is "All A are C" because if all A are B and all B are C, then it logically follows that all A are also C. This is a classic example of a syllogism, where the transitive property of inclusion is applied. The user likely skipped the question or made a mistake by not recognizing the transitive relationship between the sets A, B, and C.*

■ Q5. A shopkeeper sells a product at Rs.100, which was bought for Rs.80. What is the profit percentage?

Your Answer: __SKIPPED__
Correct Answer: 20%
■ *To find the profit percentage, we calculate the profit made (Selling Price - Cost Price) and divide it by the Cost Price, then multiply by 100. The key calculation step is: (100 - 80) / 80 * 100 = 20 / 80 * 100 = 20%. The correct answer is 20% because the profit made (Rs.20) is 20% of the cost price (Rs.80), indicating the user's answer should have been A) 20%.*

■ Q6. A seller sells a product at a 20% loss. If the selling price is Rs.800, what is the cost price?

Your Answer: __SKIPPED__
Correct Answer: Rs.1000
■ *To find the cost price, we can use the formula: Cost Price = Selling Price / (1 - Loss Percentage). Given the selling price is Rs.800 and the loss percentage is 20% or 0.20, the calculation is: Cost Price = 800 / (1 - 0.20) = 800 / 0.80 = 1000. This confirms that the correct answer is indeed Rs.1000. The user's likely mistake, if they got it wrong, would be an error in calculating the cost price using the loss percentage.*

■ Q7. What is the next number: 1, 2, 4, 7, 11, ?

Your Answer: __SKIPPED__
Correct Answer: 15
■ *The correct answer is 16, not 15. The sequence appears to be formed by adding 1, 2, 3, 4, and so on to the previous term. The key calculation step is: 1 + 1 = 2, 2 + 2 = 4, 4 + 3 = 7, 7 + 4 = 11, 11 + 5 = 16.*

■ Q8. What is the next number: 2, 6, 12, 20, 30, ?

Your Answer: __SKIPPED__
Correct Answer: 42
■ *The correct answer is 42 because the pattern of the sequence is obtained by adding 4, 6, 8, 10, ... to the previous term. The key calculation step is: 2 + 4 = 6, 6 + 6 = 12, 12 + 8 = 20, 20 + 10 = 30, 30 + 12 = 42. This sequence is formed by adding consecutive even numbers, which is why the next number in the sequence is 42.*

■ Q9. If it is true that all cats are animals and some animals are dogs, what can be concluded?

Your Answer: __SKIPPED__
Correct Answer: Some animals are not cats
■ *The correct answer is "Some animals are not cats" because we know that all cats are animals, but it's also stated that some animals are dogs, implying that not all animals are cats. This conclusion is logically derived from the given statements, as the presence of dogs among animals indicates that animals are not exclusively cats. The user's likely mistake would be incorrectly assuming an equivalence between cats and animals, or misinterpreting the statement about dogs.*

■ Q10. What is the next number: 3, 6, 12, 24, 48, ?

Your Answer: __SKIPPED__
Correct Answer: 96
■ *The correct answer is 96 because the pattern of the sequence is obtained by multiplying the previous term by 2. The key calculation step is: 48 * 2 = 96. This confirms that the next number in the sequence should indeed be 96, following the consistent doubling pattern.*

■ MCQ SECTION (0/10 - 0.0%)

■ Q1. What is the purpose of a method in Java?

Your Answer: __SKIPPED__
Correct Answer: To organize code into reusable parts
■ *The correct answer, "To organize code into reusable parts", is right because methods in Java allow for modular and reusable code, making it easier to maintain and update programs. This is achieved by breaking down a program into smaller, manageable parts, each with its own specific function. The user's skipped answer doesn't provide insight into their thought process, but a common mistake might be confusing the purpose of methods with that of variables (storing data) or exception handling mechanisms.*

■ Q2. Why is it important to test Java programs regularly?

Your Answer: __SKIPPED__
Correct Answer: To ensure the program works correctly
■ *The correct answer, "To ensure the program works correctly," is right because regular testing helps identify and fix bugs, ensuring the program functions as intended. This process saves time and resources in the long run by catching errors early on. If the user got it wrong, their likely mistake was misunderstanding the purpose of testing, which is not to add complexity or difficulty, but to validate the program's functionality.*

■ Q3. What is the benefit of using methods in Java?

Your Answer: __SKIPPED__
Correct Answer: To organize code into reusable parts
■ *The correct answer, "To organize code into reusable parts", is right because methods in Java allow for modular programming, where a large program is broken down into smaller, manageable pieces of code that can be reused throughout the program. This organization makes the code more maintainable, readable, and efficient. The user likely skipped the question or may have misunderstood the concept of methods, which are often mistakenly thought to only improve efficiency, but their primary benefit is code reusability and organization.*

■ Q4. What is the purpose of a variable in Java?

Your Answer: __SKIPPED__
Correct Answer: To store a data value
■ *The correct answer is A) To store a data value, because in Java, a variable is a named location in memory that holds a value of a specific data type. This allows developers to store, manipulate, and reuse data throughout their program. The user likely skipped this question or may have mistaken the purpose of a variable with other programming concepts, such as methods (which perform tasks) or exception handling (which deals with errors).*

■ Q5. What is the file extension for a Java file?

Your Answer: __SKIPPED__
Correct Answer: .java
■ *The correct answer is A) .java because Java source code files are typically saved with this file extension. This is where developers write and edit their Java code, and it's then compiled into a .class file. The user likely skipped the question, but if they had answered incorrectly, they might have confused .java with .class, which is the compiled form of a Java file, or .jar, which is a package file containing compiled Java classes.*

■ Q6. What is a package in Java?

Your Answer: __SKIPPED__
Correct Answer: A collection of related classes
■ *The correct answer is A) A collection of related classes, because in Java, a package is a way to organize related classes, interfaces, and other types to avoid naming conflicts and make it easier to find and use related classes. This is the fundamental concept of packaging in Java, which allows developers to group related classes together. The user's likely mistake would be confusing packages with other Java concepts, such as methods or variables, which are not the primary components of a package.*

■ Q7. What are the types of data types in Java?

Your Answer: __SKIPPED__
Correct Answer: int, double, char, String, and arrays
■ *The correct answer is B) int, double, char, String, and arrays. This is because Java has two main categories of data types: primitive types (such as int, double, and char) and reference types (such as String and arrays). The user's likely mistake was including interfaces and classes, which are not data types in Java, but rather concepts used for object-oriented programming. Note: The question does not require a mathematical calculation, so no calculation step is provided.*

■ Q8. What is Java primarily used for?

Your Answer: __SKIPPED__
Correct Answer: Web development, mobile apps, and enterprise software
■ *The correct answer is A) Web development, mobile apps, and enterprise software because Java is a versatile and widely-used programming language that is well-suited for developing large-scale applications, including web, mobile, and enterprise software. Its platform independence, strong security features, and vast ecosystem of libraries and tools make it a popular choice for these domains. Users who skipped this question or chose an incorrect option may have underestimated Java's breadth of applications or confused it with other languages that are more specialized in areas like machine learning or network security.*

■ Q9. What are the prerequisites for learning Java?

Your Answer: __SKIPPED__
Correct Answer: A computer, Java JDK, and basic knowledge of programming logic
■ *The correct answer is C) A computer, Java JDK, and basic knowledge of programming logic. This is because having a computer is necessary to run Java, the Java JDK (Java Development Kit) is required to compile and run Java programs, and basic knowledge of programming logic is essential to understand the fundamentals of Java programming. The user's likely mistake was not recognizing the necessity of the Java JDK, as it is a crucial tool for Java development, or overestimating the required level of programming knowledge.*

■ Q10. What is the purpose of compiling a Java program?

Your Answer: __SKIPPED__
Correct Answer: To convert the code into a executable file
■ *The correct answer is C) To convert the code into a executable file, because compiling a Java program translates the source code into an intermediate format called bytecode, which can be executed by the Java Virtual Machine (JVM). This step is necessary to prepare the program for execution, as the JVM can only run bytecode, not the original source code. The user's likely mistake was confusing compilation with other steps in the programming process, such as running or debugging the program.*

■ CODING SECTION (0/5 - 0.0%)

■ Q1. Write a Java program that reads a student's grades from stdin and calculates their average grade. **Input Format:** - The first line: the student's n...

Your Answer:

```
import java.util.Scanner;

public class StudentAverage {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String studentName = sc.nextLine();  // Read student's name

        double sum = 0;
        int count = 0;

        // Read grades until no more input
        while (sc.hasNextDouble()) {
            double grade = sc.nextDouble();
            sum += grade;
            count++;
```

```
        }

        sc.close();

        if (count == 0) {
            System.out.println("0");
        } else {
            double average = sum / count;
            System.out.println(average);
        }
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String name = scanner.nextLine();
        double sum = 0;
        int count = 0;

        while (scanner.hasNextLine()) {
            String[] input = scanner.nextLine().split(" ");
            if (input.length == 2) {
                try {
                    double grade = Double.parseDouble(input[1]);
                    sum += grade;
                    count++;
                } catch (NumberFormatException e) {
                    // Ignore invalid input
                }
            }
        }

        if (count == 0) {
            System.out.println("No grades provided");
        } else {
            double average = sum / count;
            System.out.println(average);
        }
    }
}
```

■ *The code failed because it doesn't correctly parse the input format, which includes a subject name before each grade. The `while (sc.hasNextDouble())` loop will throw an exception when it encounters a non-numeric subject name, causing the program to fail. This incorrect parsing results in the program not accurately calculating the average grade.*

■ Q2. Write a Java program that reads a string and an integer from stdin, applies a Caesar cipher with the given shift, and prints the encrypted string. **...

Your Answer:

```
import java.util.Scanner;

public class CaesarCipher {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String text = sc.nextLine();
        int shift = Integer.parseInt(sc.nextLine());
```

```
        sc.close();

        StringBuilder result = new StringBuilder();

        shift = shift % 26; // handle large shifts

        for (int i = 0; i < text.length(); i++) {
            char ch = text.charAt(i);

            if (Character.isUpperCase(ch)) {
                char encrypted = (char) ((ch - 'A' + shift + 26) % 26 + 'A');
                result.append(encrypted);
            }
            else if (Character.isLowerCase(ch)) {
                char encrypted = (char) ((ch - 'a' + shift + 26) % 26 + 'a');
                result.append(encrypted);
            }
            else {
                result.append(ch); // keep spaces and special characters unchanged
            }
        }

        System.out.println(result.toString());
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String inputString = scanner.nextLine();
        int shift = scanner.nextInt();
        scanner.close();

        StringBuilder encryptedString = new StringBuilder();
        for (char c : inputString.toCharArray()) {
            if (Character.isLetter(c)) {
                char base = Character.isUpperCase(c) ? 'A' : 'a';
                encryptedString.append((char) ((c - base + shift) % 26 + base));
            } else {
                encryptedString.append(c);
            }
        }

        System.out.println(encryptedString.toString());
    }
}
```

■ *There is no test case failure provided to analyze. The student's code is partially shown, but no test case results or error messages are given to determine what went wrong. Without the actual test case results, it's impossible to provide accurate feedback on the code's performance.*

■ Q3. Write a Java program that reads a string from stdin and prints the frequency of each word. **Input Format:** - A single line: the string to count wor...

Your Answer:

```
import java.util.*;

public class WordFrequency {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();
        sc.close();

        // Convert to lowercase and split by spaces
        String[] words = input.toLowerCase().split("\\s+");

        Map<String, Integer> map = new LinkedHashMap<>();

        for (String word : words) {
            if (map.containsKey(word)) {
                map.put(word, map.get(word) + 1);
            } else {
                map.put(word, 1);
            }
        }

        for (Map.Entry<String, Integer> entry : map.entrySet()) {
            System.out.println(entry.getKey() + ": " + entry.getValue());
        }
    }
}
```

Correct Answer:

```
import java.util.HashMap;
import java.util.Map;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        scanner.close();

        String[] words = input.split("\\s+");
        Map<String, Integer> frequencyMap = new HashMap<>();

        for (String word : words) {
            word = word.toLowerCase();
            frequencyMap.put(word, frequencyMap.getOrDefault(word, 0) + 1);
        }

        for (Map.Entry<String, Integer> entry : frequencyMap.entrySet()) {
            System.out.println(entry.getKey() + ": " + entry.getValue());
        }
    }
}
```

■ *There are no test case results provided to analyze the issue. Please provide the test case results to determine why the student's code failed. Once the results are available, I can provide feedback on what went wrong. The code snippet appears to be a partial implementation of a word frequency counter, but without test case results, it's unclear what the specific issue is.*

■ Q4. Write a Java program that reads two integers from stdin and calculates their greatest common divisor (GCD). **Input Format:** - The first line: the f...

Your Answer:

```
import java.util.Scanner;

public class GCDCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        int a = Integer.parseInt(sc.nextLine());
        int b = Integer.parseInt(sc.nextLine());

        sc.close();

        a = Math.abs(a);
        b = Math.abs(b);

        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }

        System.out.println(a);
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int num1 = scanner.nextInt();
        int num2 = scanner.nextInt();
        scanner.close();

        int gcd = calculateGCD(num1, num2);
        System.out.println(gcd);
    }

    public static int calculateGCD(int a, int b) {
        if (b == 0) {
            return a;
        } else {
            return calculateGCD(b, a % b);
        }
    }
}
```

■ *There is no information provided about the test case results, so it's impossible to determine what went wrong. The student's code appears to implement the Euclidean algorithm for calculating GCD, which is a correct approach. Without specific test case results, no further feedback can be provided.*

■ Q5. Write a Java program that reads an integer from stdin and prints the corresponding Fibonacci number. **Input Format:** - A single line: the index of ...

Your Answer:

```
import java.util.Scanner;

public class FibonacciNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = Integer.parseInt(sc.nextLine());
        sc.close();

        if (n < 0) {
            System.out.println("0");
            return;
```

```
        }

        if (n == 0) {
            System.out.println(0);
            return;
        }

        if (n == 1) {
            System.out.println(1);
            return;
        }

        long a = 0, b = 1;
        long result = 0;

        for (int i = 2; i <= n; i++) {
            result = a + b;
            a = b;
            b = result;
        }

        System.out.println(result);
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        scanner.close();

        long[] fib = new long[n + 1];
        fib[0] = 0;
        if (n >= 1) fib[1] = 1;
        for (int i = 2; i <= n; i++) {
            fib[i] = fib[i - 1] + fib[i - 2];
        }

        System.out.println(fib[n]);
    }
}
```

■ *There are no test case results provided to analyze the issue. Please provide the test case results to determine what went wrong with the student's code. Without the test case results, it's impossible to identify the problem. Please provide the failed test case outputs to give accurate feedback.*

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.
